SUPPORTING ADULT GUIDANCE

The rule behind the sequence

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This is a **bridging** lesson. It joins the two faces of a quadratic the learner has met so far — the growing **sequence** (Lessons 1–2) and the **expression** (Lesson 4) — which until now have sat side by side without an explicit link. The core idea is that a quadratic sequence is **generated by a quadratic expression** (its position-to-term rule): substitute the position n and the term comes out. The lesson makes this concrete by showing that the Lesson-1 dot rectangle (n by $n + 1$) **is** the expression — its count equals its area equals $n(n + 1) = n^2 + n$. It also shows a rule written two ways (brackets and expanded) generating the same sequence, which previews expanding (Lesson 4), and — as a stretch — explains **why** the second difference is 2a. Finding an unknown quadratic rule from scratch (via simultaneous equations) is intentionally left for later in the pathway.

Driving Question: *How can a sequence and an expression be the same quadratic?*

The mathematics, in plain terms

The learner has been treating quadratic **sequences** and quadratic **expressions** as if they were separate topics. They are not: a sequence is just what you get when you **substitute the position numbers into a rule**, and for a quadratic sequence that rule is a quadratic expression. The cleanest way to see it is the dot picture from Lesson 1. The rectangle at stage n has n rows and n **+ 1** columns, so the number of dots is n**(**n **+ 1)**. Cut that rectangle into an n **by** n **square** and **one extra column of** n, and the count is n^2 **+** n. So the count (the sequence), the area (the rectangle) and the expression $n^2 + n$ are three names for one quadratic. From there the learner substitutes rules to generate sequences (n = 1, 2, 3… → term), and sees that a rule in brackets, $(n + 1)(n + 2)$, and its expanded form, $n^2 + 3n + 2$, always give the same sequence — which is exactly why **expanding** (next lesson) is a fair move: it changes the look of the rule, never the rule itself. The **stretch** answers a natural question: *why* is the second difference always 2a? Because the first differences of a quadratic themselves form a **linear** sequence. For $y = an^2 + bn + c$ the first differences work out to $2an + (a + b)$ — a linear rule whose constant step is **2a** — so the second difference is 2a for the same reason a linear sequence has a constant first difference. (You do not need that algebra to support the lesson; the concrete version — 'the first differences climb by the same amount, and that amount is 2a' — is enough.)

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — build the sequence from a rule

The rule is $n(n + 1)$. Substitute the position n = 1, 2, 3, 4, 5 to generate each term.

1. $n = 1$	$1 \times 2 = 2$
2. $n = 2$	$2 \times 3 = 6$
3. $n = 3$	$3 \times 4 = 12$
4. $n = 4$	$4 \times 5 = 20$

the sequence is 2, 6, 12, 20, 30

Example 2 — one rule, two forms

$(n + 1)(n + 2)$ and $n^2 + 3n + 2$ are the same rule. Substitute n = 1 into both to check they agree.

1. the brackets, $n = 1$	$(2)(3) = 6$
2. the expanded form, $n = 1$	$1 + 3 + 2 = 6$
3. both give the same term, so	one rule, two forms

$(n + 1)(n + 2) = n^2 + 3n + 2$

Substitute the rule; read the rectangle

Two moves carry this lesson. First, **substitute**: a rule is a machine you feed the position n into, and the term drops out — do it for n = 1, 2, 3… and the sequence appears. Insisting on neat substitution (n = 3: $3^2 + 3 = 12$) makes 'the rule generates the sequence' concrete rather than abstract. Second, **read the rectangle**: the Lesson-1 dot rectangle is the expression, so keep the picture in view —

count = area = $n^2 + n$ — because it is the single strongest bridge between the number face and the symbol face. When a learner sees the brackets and the expanded form give the same sequence, name that turning one into the other is **expanding** (the next lesson's work), so the join is felt, not just stated. Keep the general 'find the rule from scratch' work for later; here the rules are given, and the aim is to *connect*, not to derive. The stretch — why the second difference is 2a — is best met concretely first (watch the first differences climb by the same amount), with the algebra reserved for a confident supporter.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The dots are the expression: grow the rectangle and read the count as $n^2 + n$.	~12 min
Movement	Rule → Sequence machine: substitute a rule to generate the sequence and its difference table.	~12 min
Reflection	Two ways to write the rule: substitute into a bracket form and an expanded form and see they match.	~10 min
Creativity	Why is the second difference 2a? (stretch): the first differences form a linear sequence climbing by 2a.	~7 min
Rest	A pause after joining a sequence and an expression into one quadratic.	~2 min
Nutrition	Mode B (intellectual): what the learning feeds — the confidence that mathematics is connected, not a list of topics.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Treating the sequence and the expression as unrelated

Repair: “The expression is the rule that makes the sequence — substitute the position n and the term comes out. Try n = 1, 2, 3.”

Writing $n \times n$ as $2n$ instead of n^2

Repair: “$n \times n$ is n^2 (the square), not $n + n$. The square part of the rectangle is the n^2 term.”

Thinking brackets and the expanded form are different rules

Repair: “Substitute the same n into both — they always give the same term, so they are the same rule written two ways.”

Substituting into only part of the rule

Repair: “Every n in the rule becomes the position number. In $n^2 + 2n$ with $n = 4$: $4^2 + 2\times4 = 16 + 8$.”

(stretch) Thinking the second difference equals a, not 2a

Repair: “Watch the first differences climb: they go up by 2a each time, not a. For $2n^2$ they climb by $4 = 2 \times 2$.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: substituting a value into a rule (Lesson 1), the constant second difference (Lesson 2), and the area of a rectangle. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner substituting and connecting:

- “Substitute n = 1, 2, 3 into the rule — what sequence appears?”
- “Draw the dot rectangle. How does the count become $n^2 + n$?”
- “Do the brackets and the expanded form give the same term? Try n = 2.”
- “Stretch: the first differences climb by how much each time? Compare it with 2a.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why introduce this between the sequence work and the expanding work?”

Because otherwise the two look unrelated. This lesson shows the link: the sequence is generated by an expression (its rule), and the dot rectangle from Lesson 1 is that expression. It gives a reason for expanding, which comes next.

“Do they need to find the rule of a sequence themselves?”

Not here — the rules are given, and the aim is to connect the sequence and the expression. Finding an unknown quadratic rule from its terms (which needs simultaneous equations) is left for later in the pathway.

“Why is the second difference exactly 2a?”

Because the first differences of a quadratic form a linear sequence, and a linear sequence climbs by the same amount every time. For $y = an^2$ that amount is 2a. The lesson shows this concretely; the algebra (first difference = 2an + (a + b)) is in the stretch.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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